



Mitarchuk Volodimir

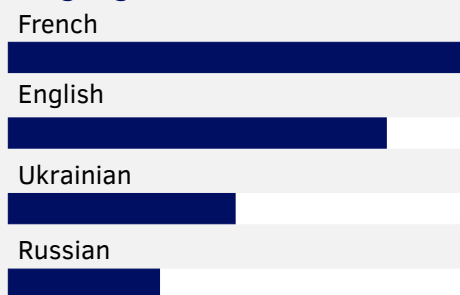
PhD Candidate in theoretical Machine Learning

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About me

My background and research experience lie at the interface between fundamental mathematics and theoretical computer science. My research is a continuation of my specialization in the Mathematical Foundations of Machine Learning. I am currently part of the INRIA MALICE research team, which aims to integrate physical knowledge within machine learning frameworks.

Languages skill



Scale: 0 (basic skills) - 6 (expert).

A short Biography

My name is Volodimir, and I am native of L'viv in Ukraine. I pursued a classic university cursus of studying mathematics i.e. Bachelors/Masters in fundamental mathematics, with a specialization in Mathematical Foundations of Machine Learning at the university ENS (*École Normale Supérieure*) at Lyon France. After my Master degree I started a PhD in theoretical computer science. My thesis aim is to study recurrent neural networks from a theoretical standpoint (Generalization, Learnability and Expressivity). Currently I hold a post doc position at the MALICE inria team, and my research interest is the application of Neural Operators to time series.

Education

2021-2025	Ph.D. Computer Science Duration: October 2021 - January 2025. Title: "Saturation in Recurrent Neural Networks: Expressivity, Learnability and Generalization". Supervisors: Rémi Eyraud, Amaury Habrard and Rémi Emonet.	University Jean Monnet
2020-2021	Mathematics Second year of masters degree in advanced mathematics, specializing in the mathematical foundations of machine learning	Ecole Normale Supérieure de Lyon
2019-2020	Mathematics First year of masters degree in fundamental mathematics.	University Aix-Marseille
2018-2019	Mathematics Third year of Mathematics bachelor's degree.	University Lille 1
2016-2018	Mathematics First and second year of Mathematics Bachelor's degree.	University D'Artois
2015-2016	Applied Mathematics and Computer Science First year of a Bachelor's degree in Mathematics and Computer Science applied to the humanities and social sciences.	University Lille 3
2010-2012	Advanced Technician's Certificate Mechanics and automated systems	High School Baggio
2007-2010	Lycée Jean Prouvé General baccalaureat engineering specialty	High School

Experiences

2025	Post Doc position Application of Neural Operators/Koopman Operator to time series. MALICE Inria team web site https://labhc-malice.github.io/	MALICE Inria team
2024	Internship supervision Topic "On the difficulty of training Saturated RNN"	First year MLDM Master
2024	TAUDoS TAUDoS https://remieyraud.github.io/TAUDoS/ a consortium research project. Launch and animation of a newsletter, co-animation of yearly workshops.	Consortium animation
2023	Research visit A 1-month visit at the MILA center in Montreal, at the team of Guillaume Rabusseau.	MILA center
2023	Reviewing LearnAur2023, NeurIPS 2023	
2022	Spring School EPIT 2022 at CIRM Marseille	Theoretical Computer Science

The main passion

I am a father since February 2021 of a small girl. Being a dad is a job/passion that deserves its own section.

Teaching

2021-2023	Introduction to Python	Bachelor
	Python practical sessions to the second year of the Physics/Chemistry bachelor's degree.	
2023-2025	Constrained Optimization	Master
	Constrained optimization math course in the MLDM (Machine Learning and Data Mining), an international master's program.	

Publications

2024	Journal	TMLR
	On the theoretical limit of gradient descent for Simple Recurrent Neural Networks with finite precision	
2024	Workshop	LearnAut2024
	Presented a work on theoretical study of incremental counting in LSTM in finite precision at the LearnAut2024 workshop at ICALP 2024 conference	
2024	Conference	AISTats
	A PAC-Bayes bound for Recurrent Neural Nets (RNN) that does not depend on the data length.	
2022	Workshop	LearnAut2022
	"Theoretical limitations of gradient descent in finite precision" at the LearnAut2022 workshop at ICALP 2022 conference	

Hobbys

Music	I produce myself live, singing alone with my guitar, at the local rock scene.
Sports	I am practicing several times a week at the gymnasium.
DIY	I enjoy building stuff with my hands (pizza oven, car maintenance).
Transport	I am passionate about the automotive design/history.